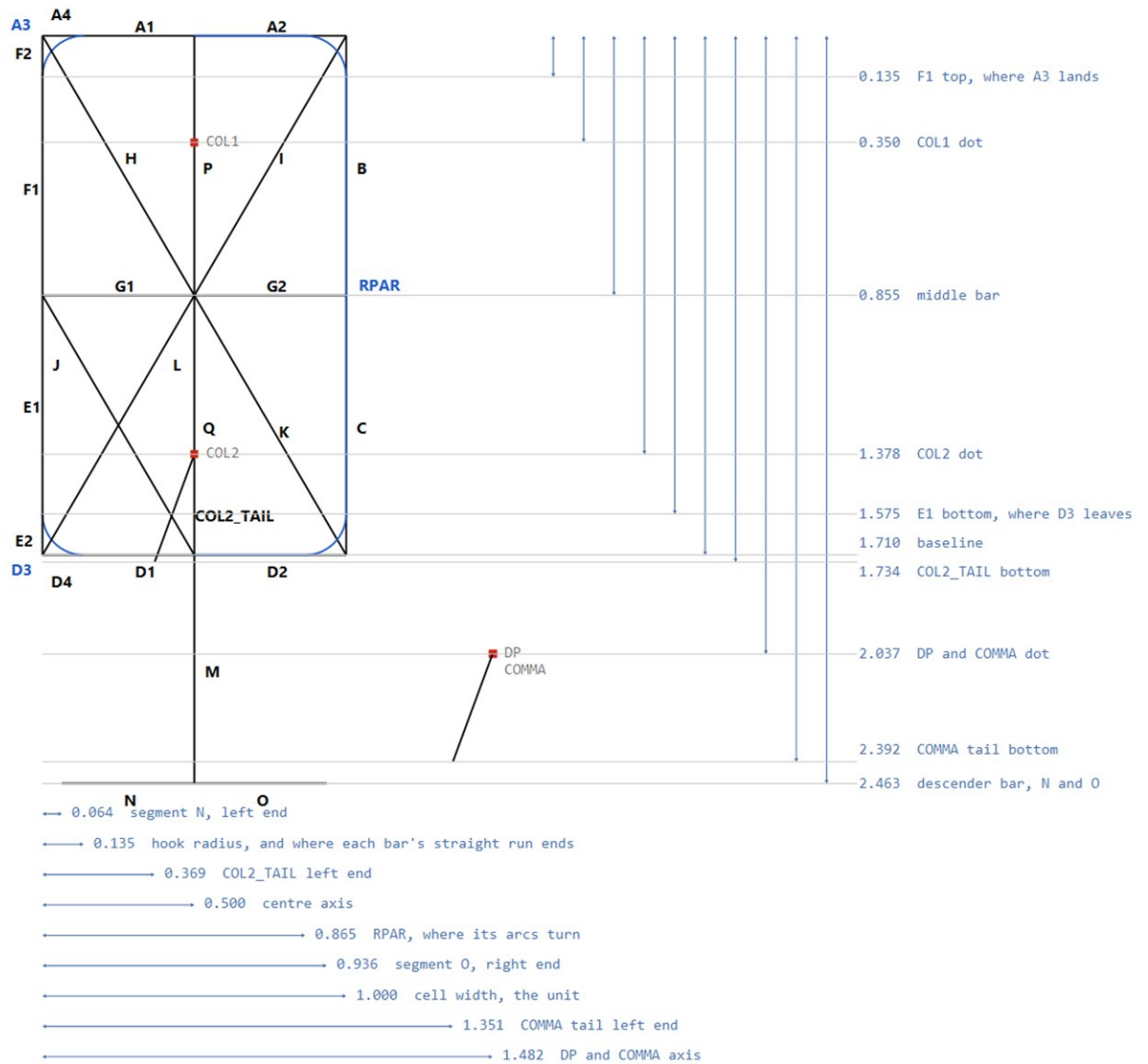


TalkRPN cell - centreline skeleton

Inspired by the HP-01 (1977) and the Litronix DL-3422, and identical to neither. The HP-01's look - hooked corners, the slant, the proportions - carried onto roughly the DL-3422's segment count, so the display can show letters as well as digits. Departures from both are deliberate: the corners hook where the DL-3422's are square, the segments meet flush where a real part leaves gaps, the decimal point sits between cells rather than taking one of its own, and there are elements here that neither part had. It is a period style, not a reproduction of any real component.

THE UNIT: segment E/F to segment B/C is 1 - the left column to the right column, centre to centre. Every length here is in that unit, across and down alike, so the centre axis is exactly 0.5 and pitch minus 1 is the clearance between neighbouring cells.

No stroke width, no slant: these are centrelines, and both are applied at render time. Origin is the top-left centreline corner; x right, y down. 28 bars + COL1 + COL2 + DP + COMMA = 32 elements; the mask is a long. Blue: A3/D3 are ALTERNATIVES to A4/D4; RPAR is the whole right paren. TalkRpnFont.kt is the source of truth - this sheet mirrors it.



TalkRPN cell - centreline listing

SEGMENT	CENTRELINES	from	to	
A1	line	0.500, 0.000	-> 0.135, 0.000	
A2	line	0.500, 0.000	-> 1.000, 0.000	
A4	line	0.135, 0.000	-> 0.000, 0.000	
A3	arc	0.135, 0.000	-> 0.000, 0.135	arc R0.135 centre 0.135,0.135
B	line	1.000, 0.000	-> 1.000, 0.855	
C	line	1.000, 0.855	-> 1.000, 1.710	
D1	line	0.500, 1.710	-> 0.135, 1.710	
D2	line	0.500, 1.710	-> 1.000, 1.710	
D4	line	0.135, 1.710	-> 0.000, 1.710	
D3	arc	0.135, 1.710	-> 0.000, 1.575	arc R0.135 centre 0.135,1.575
F2	line	0.000, 0.000	-> 0.000, 0.135	
F1	line	0.000, 0.135	-> 0.000, 0.855	
E1	line	0.000, 0.855	-> 0.000, 1.575	
E2	line	0.000, 1.575	-> 0.000, 1.710	
G1	line	0.000, 0.855	-> 0.500, 0.855	
G2	line	0.500, 0.855	-> 1.000, 0.855	
H	line	0.000, 0.000	-> 0.500, 0.855	
I	line	1.000, 0.000	-> 0.500, 0.855	
J	line	0.000, 0.855	-> 0.500, 1.710	
L	line	0.500, 0.855	-> 0.000, 1.710	
K	line	0.500, 0.855	-> 1.000, 1.710	
P	line	0.500, 0.000	-> 0.500, 0.855	
Q	line	0.500, 0.855	-> 0.500, 1.710	
COL2_TAIL	line	0.500, 1.378	-> 0.369, 1.734	
M	line	0.500, 1.710	-> 0.500, 2.463	
N	line	0.064, 2.463	-> 0.500, 2.463	
O	line	0.500, 2.463	-> 0.936, 2.463	
RPAR	compound	0.500,0.000	-> arc -> 1.000, 0.135..1.575 -> arc -> 0.500,1.710	the whole right paren

DOT CENTRES

COL1	0.500, 0.350	upper colon dot; also dots i and j
COL2	0.500, 1.378	lower colon dot, colon only
DP	1.482, 2.037	decimal point, outside the cell to the right
COMMA	1.482, 2.037	same centre, plus a tail down and left

SQUARE, side = 2 x STROKE = 0.295 - a real HP-55's decimal point is a square die

CENTRELINE BOX 1.000 wide x 1.710 tall aspect 0.585
 2.463 tall including the descender
 PITCH 2.430 origin to origin; minus 1 IS the clearance between cells
 VPITCH floor is 2.610, the ink height - below that descenders reach the next row
 SLANT 7.5 degrees, applied at render

RELATIONSHIPS

hook radius = hook start x = 0.135, so each arc is tangent to both bar and column
 F1 top = 0.135 = where A3 lands. E1 bottom = 1.575 = where D3 lands.
 F2 and E2 are the stubs that carry the left column out to the corner when it is square.
 B meets C at y = 0.855: the right column is undivided at the corners, which is what makes a 4 lopsided.
 N and O are inset 0.0640 and 0.0641 from the columns - symmetric to within measurement.
 COL2_TAIL and the COMMA tail are the SAME tail, 0.355 down and 0.131 left, hung off two different dots.
 That is what makes a semicolon a colon whose lower dot grew a tail, at no cost in new geometry.

THE THREE LEFT-COLUMN ENDINGS

square	A4 + F2	most letters
hooked	A3 + F1	0 2 3 5 7 8 9, A C G O Q S, (& and @ (F2 dark)
short	F1 alone	digit 4, no A at all - left side sits 0.135 lower than the right

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
2																
	sp	! P COL2	" F2 F1 P	# A4 A1 A2 G1 G2 N O P Q H	\$ A3 A1 A2 F1 G1 G2 C D4 D1 D2 P Q H	% A3 A1 F1 G1 P I L G2 C D2 Q	& A4 A1 P H G1 E1 G3 D1 D2 C K	' I	(A1 A3 F1 E1 D3 D1	) RPAR	* H I L K P Q G1 G2	+ G1 G2 P Q	, COMMA	- G1 G2	. DP	/ I L
3																
	0 A3 A1 A2 F1 B E1 C G3 D1 D2	1 B C	2 A3 A1 A2 B G1 G2 E1 D3 D1 D2	3 A3 A1 A2 B G1 G2 C D3 D1 D2	4 F1 G1 G2 B C	5 A3 A1 A2 F1 G1 G2 C D3 D1 D2	6 A4 A1 A2 F2 F3 G1 G2 E1 E2 C D4 D1 D2	7 A3 A1 A2 B C	8 A3 A1 A2 F1 B G1 G2 E1 C D3 D1 D2	9 A3 A1 A2 F1 B G1 G2 C D3 D1 D2	: COL1 COL2	; COL1 COL2 COL3_TAIL	< I K	= G1 D4 D1	> H L	? A4 A1 A2 B G2 Q
4																
	a A3 A1 A2 B C D3 D3 D2 G1 G1 G1 Q	A A3 A1 A2 F1 B G1 G2 E1 E2 C	b A4 A1 A2 B C D4 D1 D2 G2 G2 P Q	c A3 A1 A2 F1 E3 G3 D1 D2	d A4 A1 A2 B C D4 D1 D2 G2 P Q	e A4 A1 A2 F2 F3 G1 E1 E2 D4 D3 D2	f A4 A1 A2 F2 F3 G1 E1 E2	g A3 A1 A2 F1 E1 G3 D3 D2 C G2	h F2 F1 B G1 G2 E1 E2 C	i A4 A1 A2 P Q D4 D1 D2	j B C E3 D3 D1 D2	k F2 F1 E1 E2 G3 I K	l F2 F1 E3 E2 D4 D1 D2	m F2 F1 H I B E1 E2 C	n F2 F1 H B K E1 E2 C	o A3 A1 A2 F1 B E1 C D3 D1 D2
5																
	P A4 A1 A2 F2 F1 B G1 G2 E1 E2	Q A3 A1 A2 F1 B E1 C D3 D1 D2 K	R A4 A1 A2 F2 F1 B G1 G2 E1 E2 K	S A3 A1 A2 F1 G1 G2 C D4 D1 D2	T A4 A1 A2 P Q	U F2 F1 B E1 C D3 D1 D2	V F2 F1 E1 E2 I L	W F2 F1 B E1 E2 C L K	X H I L K	Y H I Q	Z A4 A1 A2 I L D4 D3 D2	[A2 P Q H O	\ H K	] A1 P Q H N	^ L K	_ D4 D1 D2
6																
	\ H	a G1 Q D4 D1 L	b F2 F1 E1 G1 D3 D1 Q	c E1 G1 D3 D1	d E1 G1 D3 D1 P Q	e E1 E2 G1 D4 D1 L	f A2 P Q G1 G2	g E1 G1 D3 D1 Q H N	h F2 F1 E1 E2 G1 Q	i Q COL1	j Q H N COL1	k F2 F1 E1 E2 G1 J	l P Q	m G1 G2 E1 E2 C Q	n E1 E2 P Q	o E1 G1 D3 D1 Q
7																
	p C D2 D2 Q H	q E1 G1 D3 D1 Q H O	r E1 E2 G1	s G1 J D4 D1	t P Q G1 G2 D2	u E1 D3 D1 Q	v E1 E2 L	w E1 E2 C L K	x P L	y E1 E2 D4 D1 Q H N	z G1 L D4 D1	{ A2 P G1 Q H O	P Q H	} A1 P G2 Q H N	~ F3 A3 A1 P G2 B	"